Fundamentals of AI & ML (CSA2001) – BYOP
Project Report

Personal Fitness Recommender (CLI)

Submitted by

Aviral Sahu

Registration No.: 25BAI10409

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1 Introduction and Motivation

Many students want to get fit but do not know how to plan workouts according to their goal and fitness level. Most online plans are generic, and people also forget what they did last time because they do not track anything.

I wanted to build a small tool that:

- suggests routines based on the user's **goal** (build muscle / lose weight / general fitness) and **fitness level** (beginner / intermediate / advanced),
- runs fully in the **terminal**, so it is easy to use on any laptop, and
- lets the user **log workouts** and see simple progress and statistics.

This led to the idea of the **Personal Fitness Recommender**, a command-line interface (CLI) application that combines a simple machine learning model with rule-based logic.

2 Problem Statement and Objectives

Problem Statement

Design a simple AI-based system that can:

1. Capture a user's basic details (age, weight, height, fitness level, goal).
2. Recommend a suitable workout style using a machine learning model.
3. Generate a detailed workout routine (with home and gym variations).
4. Log each workout and show history plus basic analytics.

Objectives

The main objectives are:

- Implement registration, login and logout in a CLI.
- Use a **Decision Tree Classifier** to select a routine type.
- Store users and workouts in JSON files so data is persistent.
- Provide features for:
  - suggesting routines,
  - logging workouts,
  - viewing progress,
  - viewing a workout summary (total sessions, minutes, breakdowns).

3 Tools and Requirements

Software and Libraries

- **Language**: Python 3.8+
- **Libraries**: `scikit-learn`, `pandas`, `numpy`, and standard Python libraries such as `json`, `os`, `datetime`.
- **Platform**: Works on Windows / Linux / macOS in any terminal.

Data Storage

- `users.json` – stores user profiles (username, password, age, weight, height, fitness level, goal).
- `workouts.json` – stores workout logs (date, goal, location, description, duration).

4 System Design

The project is divided into three main Python files.

data.py – Data Manager

Responsibilities:

- Handle all file operations for users and workouts.
- Key functions:
  - `register_user`, `login_user`, `get_user_data`, `update_goal`
  - `log_workout` – saves date, goal, location, description, duration
  - `get_progress` – returns all workouts for a user

model.py – Machine Learning Model

Responsibilities:

- Create a small synthetic dataset with:
  - age, weight, height, fitness_level, goal,
  - routine type: `Cardio Focus`, `Strength Training`, `Balanced`.
- Encode categorical features manually (e.g., beginner → 0, intermediate → 1, advanced → 2).
- Train a **DecisionTreeClassifier** and save it as `fitness_model.pkl`.
- Provide `predict_routine(user_data)` to return one of the three routine types.

main.py – CLI Application

Responsibilities:

- Implement the `FitnessCLI` class and the interactive menu.
- Support:
  - **Register**: create user profile with validation.
  - **Login / Logout**: manage current logged-in user.
  - **Set Goal**: update the goal for the user.
  - **Suggest Routine**:
    - load user profile,
    - call the ML model for routine type,
    - print detailed home and gym routines based on goal and level.
  - **Log Workout**: ask for session goal, location, description and duration; store via `DataManager`.
  - **View Progress**: print all workouts for that user.
  - **Workout Summary**: show total sessions, total minutes, and breakdown by goal and location.

5 Algorithm Used: Decision Tree

The AI part uses a **Decision Tree Classifier** from scikit-learn.

Inputs

- age, weight, height (numeric),
- encoded fitness level (0, 1, 2),
- encoded goal (0, 1, 2).

Output

- routine type: `Cardio Focus`, `Strength Training`, or `Balanced`.

Training Steps

1. Build a small `pandas` DataFrame with sample profiles and labels.
2. Map string categories to integers (manual encoding).
3. Split into train and test sets.
4. Train the decision tree and save it using `joblib`.

For a logged-in user, the app forms a single-row DataFrame with the profile, calls `model.predict`, and then uses rule-based templates to generate the actual list of exercises.

6 Outputs (Screenshots)

This section shows basic outputs from the Personal Fitness Recommender CLI such as the main menu and registration flow.

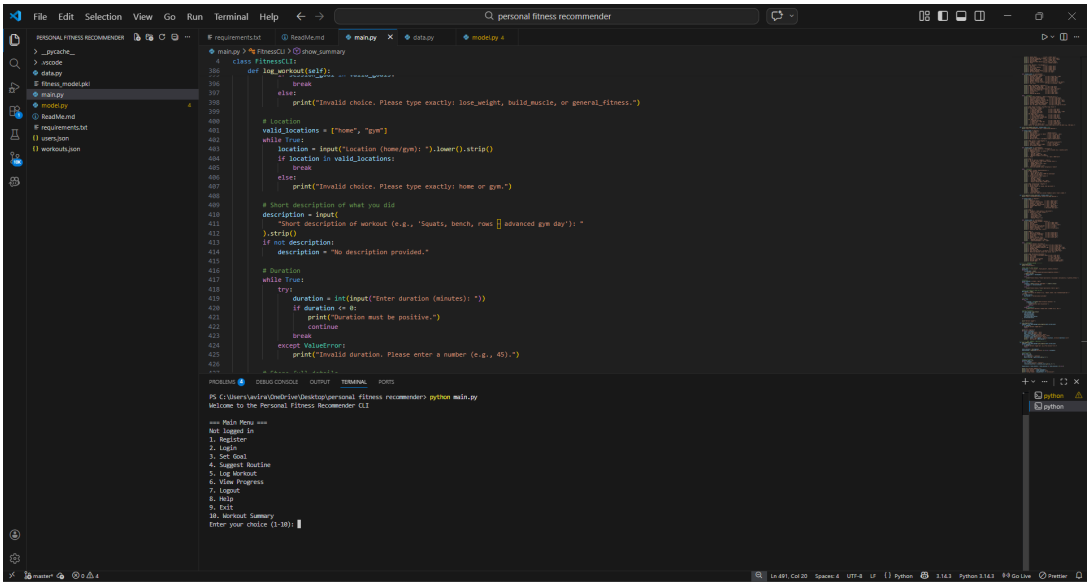


Figure 1: Main menu showing all options (Register, Login, Suggest Routine, Log Workout, View Progress, Workout Summary, etc.).

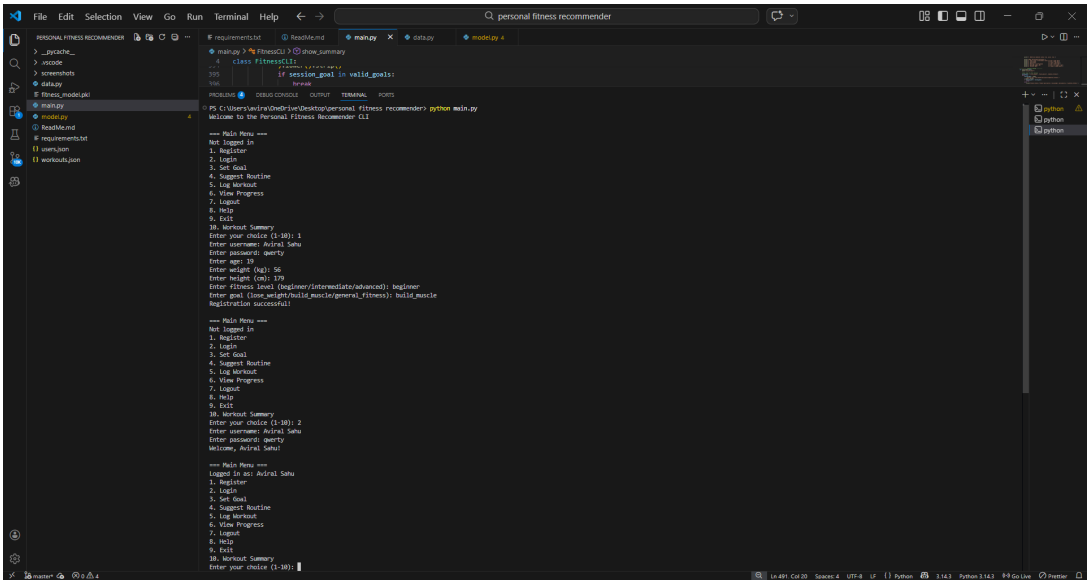


Figure 2: Registration of a new user and successful login.

7 Testing and Observations

Testing was done manually by running different flows through the CLI:

- Trying invalid inputs (wrong fitness level, text instead of numbers) to check validation and error messages.
- Creating different user types such as:
  - beginner + lose_weight,
  - intermediate + general_fitness,
  - advanced + build_muscle.
- Logging several workouts with different goals and locations.
- Checking that `view_progress` and `show_summary` match the logged data.

The following screenshots illustrate the AI routine recommendation and logging steps used during testing.

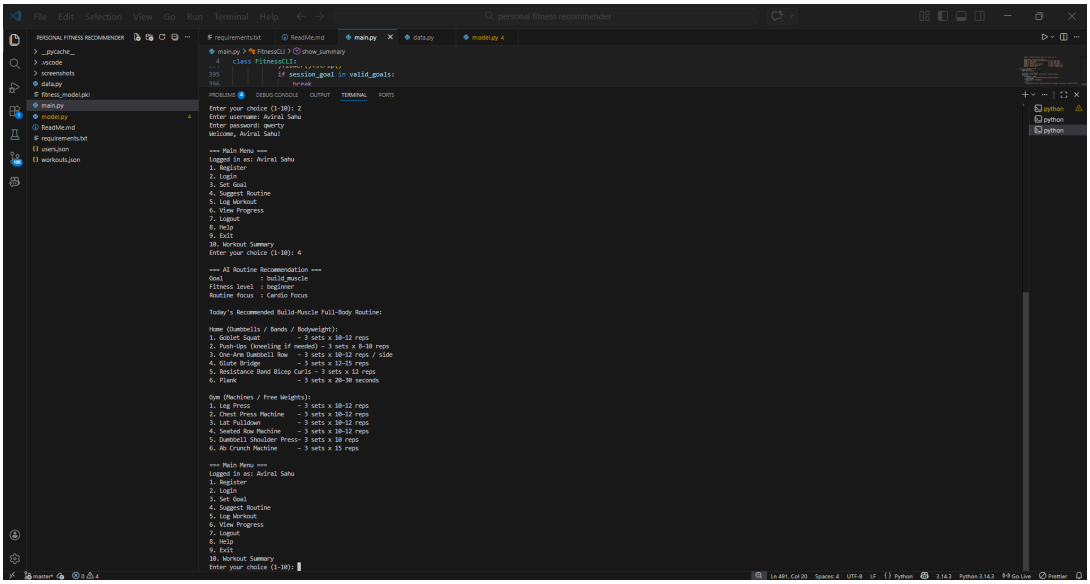


Figure 3: AI-recommended routine for an advanced user with build_muscle goal.

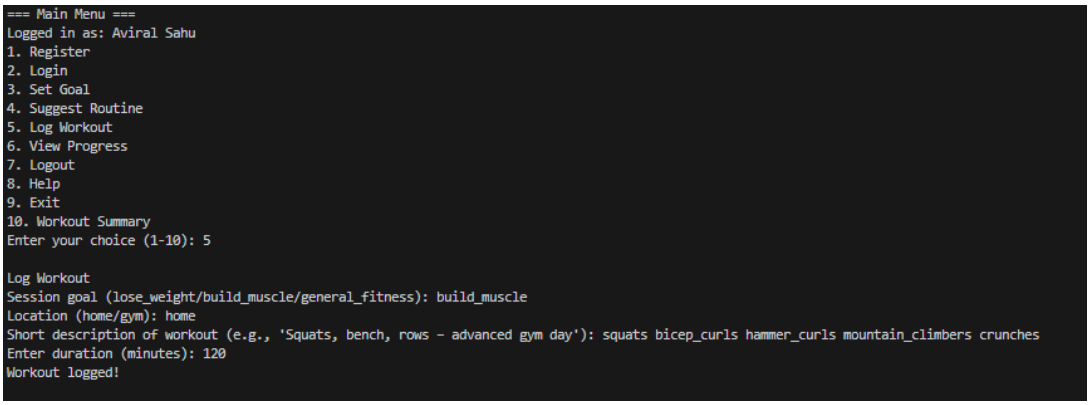


Figure 4: Logging a workout with goal, location, description and duration.

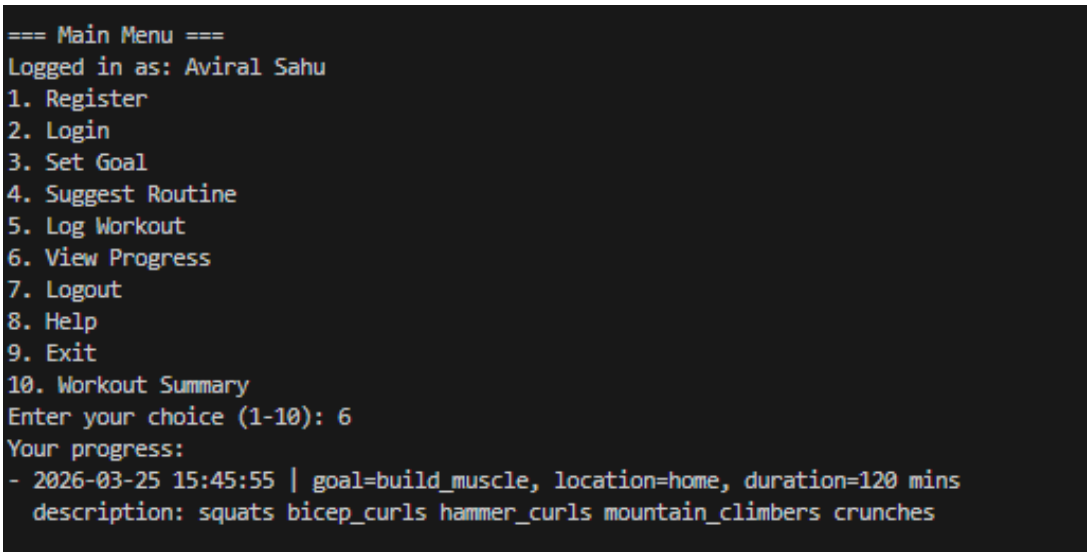
During testing, issues like label encoding errors and login state problems were found and fixed. The final version handles invalid inputs more gracefully and prints more realistic routines.

8 Results and Significance

The final system can:

- store multiple user profiles with different goals and levels,
- recommend detailed workout routines that vary by fitness level and environment (home/gym),
- log workouts with goal, location, description and duration, and
- provide a quick summary of total sessions, total minutes, and how often each goal and location was used.

The next two screenshots show the logged workouts and the workout summary which support these results.



```
=== Main Menu ===
Logged in as: Aviral Sahu
1. Register
2. Login
3. Set Goal
4. Suggest Routine
5. Log Workout
6. View Progress
7. Logout
8. Help
9. Exit
10. Workout Summary
Enter your choice (1-10): 6
Your progress:
- 2026-03-25 15:45:55 | goal=build_muscle, location=home, duration=120 mins
  description: squats bicep_curls hammer_curls mountain_climbers crunches
```

Figure 5: Viewing progress: list of previously logged workouts.

```
=== Main Menu ===
Logged in as: Aviral Sahu
1. Register
2. Login
3. Set Goal
4. Suggest Routine
5. Log Workout
6. View Progress
7. Logout
8. Help
9. Exit
10. Workout Summary
Enter your choice (1-10): 10

=== Workout Summary ===
Total sessions : 1
Total minutes  : 120 min
Average length : 120.0 min/session

Sessions by goal:
- build_muscle: 1 session(s)

Sessions by location:
- home: 1 session(s)
```

Figure 6: Workout summary showing total sessions, total minutes and breakdown by goal and location.

For a BYOP project, this shows how a basic machine learning model plus simple rules can be turned into a useful application, not just a notebook experiment.

9 Limitations and Future Work

Limitations

- The dataset for training the decision tree is small and synthetic, so the recommendations are not medically accurate.
- The app does not consider injuries, medical conditions, or very specific training needs.
- There is no automatic progression in sets or weights; the user must adjust manually.
- All data is stored locally; there is no cloud backup or sync between devices.

Future Enhancements

- Collect a larger, more realistic dataset for training.
- Add weekly/monthly summaries and charts (e.g., minutes per week).
- Implement progression rules that automatically increase difficulty over time.
- Export logs to CSV and build visual graphs using libraries like `matplotlib`.
- Build a simple web or mobile front-end on top of the same logic.

10 Conclusion

The **Personal Fitness Recommender (CLI)** meets the main goals of the project. It uses a **Decision Tree Classifier** to choose a routine style, generates personalized workouts based on fitness level and goal, and allows users to log workouts and view both detailed history and simple analytics.

Through this project, I (Aviral Sahu, Reg. No. 25BAI10409) practised:

- applying a supervised learning algorithm to a practical problem,
- designing a complete CLI application with user accounts and persistent storage,
- debugging, validating inputs and refining features based on testing.

The project is small but genuinely useful for my own training, and it can be expanded in many directions in the future.